SPACE TECHNOLOGY · ARTIFICIAL INTELLIGENCE

Artificial Intelligence in Space Technology

“How do we explore places that are too distant, dangerous or unknown for humans to control directly?”

Communication delays — instructions cannot always arrive in time

Limited resources — power, computing, storage and bandwidth

Too much data — more than people can inspect by hand

Dangerous places — radiation, extreme cold, unknown terrain, caves

Little human help — nobody can correct every mistake live

AI helps spacecraft and robots interpret data, make decisions, navigate hazards and prioritise useful observations with less continuous human intervention.



What can AI do in space that humans cannot easily do?

01

Autonomous navigation

A rover detects hazards and plans a safe path while driving.

TECHNIQUES
Computer vision, 3D mapping, path planning

02

Onboard data filtering

A satellite spots useful images before sending them to Earth.

TECHNIQUES
CNNs, classification

03

Large-scale pattern detection

Telescope datasets are scanned for possible exoplanets and other signals.

TECHNIQUES
Deep learning, classification

04

Prediction and anomaly detection

Patterns in solar data or sensor readings may forecast events or flag faults.

TECHNIQUES
Predictive ML, anomaly detection

05

Reaching inaccessible places

Robots map dark, unmapped caves and make local decisions.

TECHNIQUES
SLAM, probabilistic reasoning, sensor fusion, planning

AI is useful because it can do these tasks on its own, continuously, or at a scale and speed that are hard for people — especially when communication and access are limited. Not every system uses deep learning: AutoNav and cave autonomy combine perception, mapping and planning.

From Orbit to the Martian Surface

ESA Φsat-1

AI on a satellite · operational demo

PROBLEM
Cloudy images waste limited downlink bandwidth.

TECHNIQUE
CNN cloud detection running onboard.

WORKING
A hyperspectral imager captures scenes; the onboard processor labels cloudy vs clear before downlink.

RESULT
Launched 3 Sep 2020; ESA calls it the first demonstration of AI for Earth observation, filtering out poor images.

LIMIT
Small power and compute budget; a wrong call could discard useful data.

Perseverance AutoNav

NASA self-driving rover · operational

PROBLEM
Signal delay rules out joystick driving on Mars.

TECHNIQUE
Stereo vision, 3D terrain maps, hazard detection, path planning, visual odometry.

WORKING
Maps terrain ahead, routes around hazards, checks its own movement.

RESULT
NASA: up to ~120 m/h possible, vs ~20 m/h for Curiosity's earlier AutoNav.

LIMIT
Humans still plan routes. Not designed for dark caves.

ExoMiner

Deep learning for exoplanets · research result

PROBLEM
Kepler data holds many planet-like false positives.

TECHNIQUE
Deep neural network trained on confirmed planets and false positives.

WORKING
Scores candidate signals; scientists follow up and validate.

RESULT
NASA (Nov 2021): 301 new planets validated in Kepler data.

LIMIT
Depends on data quality; a candidate is not a confirmed planet.

More applications: the Sun, spacecraft health and the future

Solar flare prediction

Researched application

PROBLEM
Flares can disrupt radio links and satellites.

APPROACH
ML models learn patterns in solar magnetic-field and image data.

BENEFIT
Earlier warning for operators and planners.

LIMIT
Skill varies by model and forecast window; false alarms and misses remain.

Spacecraft failure prediction

Researched application area

PROBLEM
Faults are hard or impossible to repair after launch.

APPROACH
Anomaly detection on telemetry: temperature, voltage, current.

BENEFIT
Earlier investigation and better maintenance planning.

LIMIT
Real failures are rare; unusual ≠ failing. No mission result claimed here.

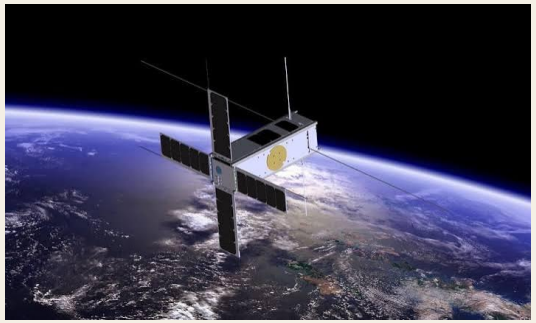
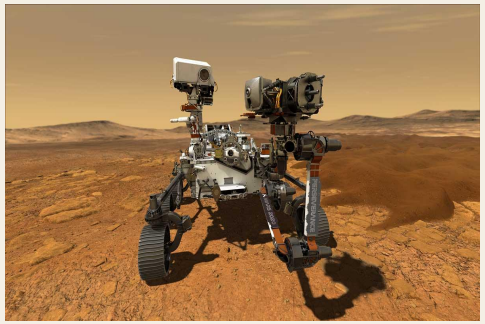
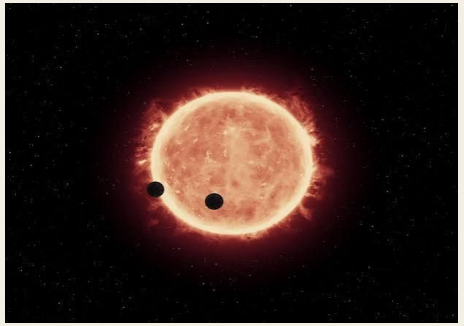
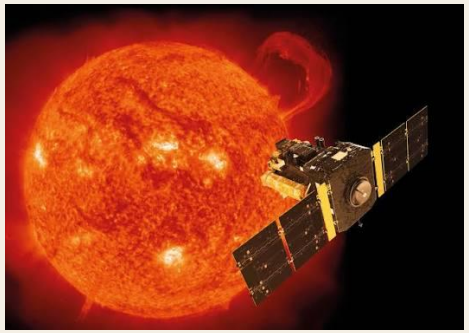
Future autonomous exploration

Proposed Ideas

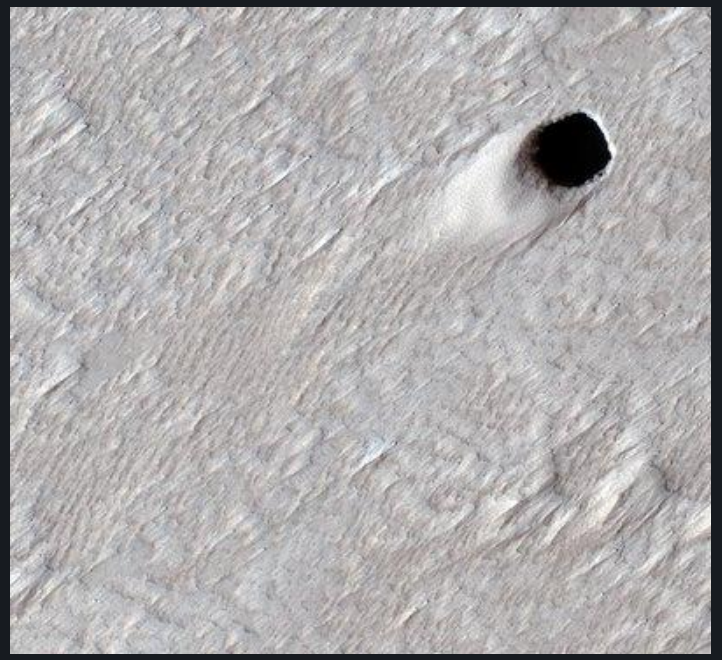
IDEA
The same principles could extend to lunar caves and Martian lava tubes.

WHAT AI ADDS
Map unfamiliar surroundings, choose safe moves, prioritise tasks.

CONSTRAINT
Communication is intermittent, so decisions must be local.


Could AI robots explore the hidden caves of Mars?



Observed on Mars: a possible lava-tube skylight.

WHY LAVA TUBES?

More protected than the surface; may hold clues to past habitability, minerals, water ice, organic molecules, or even primitive life.

But: dark, rugged, unmapped, delayed or blocked comms; wheeled rovers struggle to get in.

WHAT IS MELT? (PROPOSED CONCEPT)

Martian Exploration of Lava Tubes: a mission concept by Di et al. (2024) using ReachBot.
A support rover deploys a robot with extendable booms and grippers to inspect cave walls.

What was tested ?

Boom deployment
Rock detection and target selection
Gripping strength

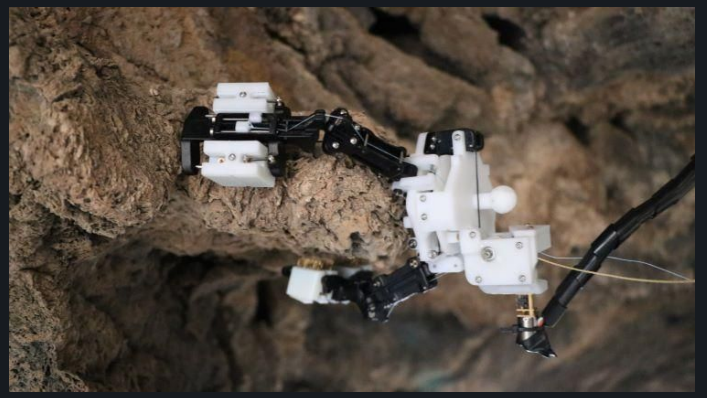
The Ultimate Goal

Explore difficult-to-access caves and lava tubes on Mars, which conventional wheeled rovers may struggle to traverse.

Result

The prototype successfully targeted and grasped multiple rock features. The reported tests measured grasp pull-off forces of at least 34.2 N for the tested candidate sites





Plausible in principle, but the full mission has not flown. Some parts were tested in Earth lava-tube analogs.

Tested At

Lavic Lake volcanic field, Mojave Desert, California.

Can a robot map and explore a cave without a human driver?



Sense Cameras and depth/range sensors observe the cave.

Map SLAM: build a map and estimate position at once.

Assess hazards Estimate traversability, allowing for uncertainty.

Decide Pick where to explore next without per-step commands.

Report Flag areas of interest for the science team.

What was tested ?

- Two Spot quadruped robots with JPL's NeBula autonomy software
- 22 tests conducted over three days
- Compared one vs. two robots and high vs. low autonomy
- Evaluated exploration efficiency and detection of 7 artificial science targets

Test Results

- Autonomous exploration worked
- They mapped the cave and searched for targets
- Two robots were generally more effective

TESTED AT

Lava Beds National Monument, California, USA
May 2022 — Earth-based Martian-cave analog experiments
Valentine Cave: dark caves, no full prior map, no GPS.

NeBula = Networked Belief-aware Perceptual Autonomy.

What has AI achieved — and what still needs work?

DEMONSTRATED

- Φsat-1: onboard AI cloud filtering in orbit
- AutoNav: faster autonomous driving on Mars
- ExoMiner: deep learning helped validate 301 Kepler planets
- Earth cave tests: substantial autonomous mapping and exploration
- ReachBot: grasping with a partial single-boom prototype (Mojave lava tube)

POTENTIAL IMPACT

- Better use of satellite bandwidth
- Faster exploration, fewer continuous commands
- Quicker analysis of large datasets; earlier fault and space-weather warnings
- Access to caves that may hold geological or biological clues

LIMITATIONS

- Strict energy, mass and computing limits
- Predictions and perception can be wrong
- Training data may not match every real environment
- Delays and outages complicate intervention
- Earth caves cannot match Martian gravity, atmosphere, dust or cold
- MELT/ReachBot remains proposed; autonomous science target selection needs more work

AI does not eliminate human expertise. It extends our ability to explore, observe and make decisions when direct human control is limited.